WEB DESIGN FOR EVERYBODY · CAPSTONE (TEMPLATE TRACK)

Table Tennis Central Club

Project report by Le Bao Hung

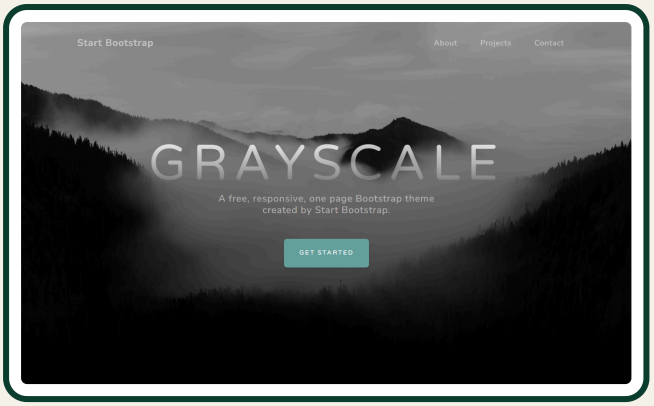
1. Overview

Table Tennis Central Club is a one-page website for a community table tennis club. It introduces the club, describes three programs (League Nights, Beginner Clinics and the Junior Academy), lets visitors request a free trial session, and lists contact details and opening hours.

The site is built on **Grayscale 7.0.6**, a free Bootstrap 5 theme by Start Bootstrap released under the MIT License. I kept the template's Bootstrap grid, navigation behaviour and section layout, and replaced or reworked everything else: content, artwork, colour theme, the sign-up form, and every accessibility and validation problem the original shipped with.

> **Live site:** https://lebaohungk05.github.io/table-tennis-basics/template-site/
>
> **Original template, unmodified (for comparison):** https://lebaohungk05.github.io/table-tennis-basics/template-original/
>
> **Source code:** github.com/lebaohungk05/table-tennis-basics/template-site
>
> **Template:** Start Bootstrap Grayscale (MIT License, kept in the project as `LICENSE`)

ORIGINAL TEMPLATE



Grayscale as downloaded

MY SITE

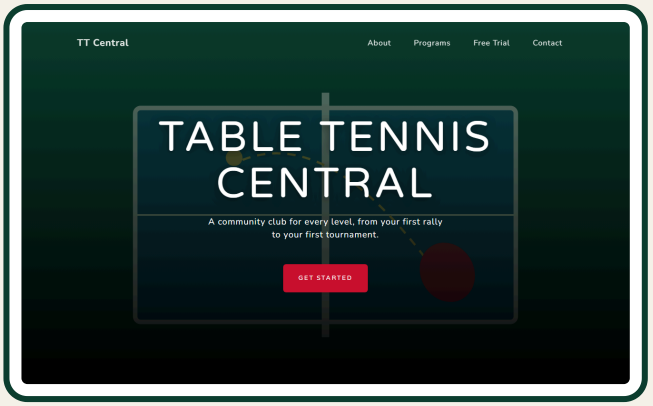


Table Tennis Central Club

2. Changes From the Original Template

What I changed, file by file

Area	Original template	My version
Content	Placeholder text (“Shoreline”, “Misty”, “Your Website”) and an empty meta description	All copy rewritten for the club, real meta description and author, sections renamed to About, Programs, Free Trial and Contact
Images	Four stock photos and an iPad mock-up, every one with `alt="..."`	Six original SVG illustrations (masthead, club hall, three programs, logo/favicon), each with descriptive alt text
Colour theme	Teal `#64a19d` on grey and black	New palette in a separate `css/custom.css` built on CSS custom properties: court green, table blue, paddle red, ball yellow
Masthead title	Gradient text that fades to fully transparent at the bottom of each letter	Solid white text with a soft shadow, sized with `clamp()` so the longer title fits every screen

Area	Original template	My version
Sign-up form	Depends on the paid SB Forms service; without an API token the button stays disabled and nothing works	Removed SB Forms; wrote `js/signup.js` with my own validation and spoken feedback (see Extra 1)
Contact cards	Fake address, a `#!` email link, a phone number	Club address, a working `mailto:` link, opening hours
Social links	Three icon-only links to `#!` (Twitter, Facebook, GitHub)	Two real links (project source and the club's guide to the sport), each with an `aria-label`
`css/styles.css`	A LESS function `fade(#fff, 80%)` leaked into the compiled CSS, which is invalid	Replaced with `rgba(255, 255, 255, 0.8)`. This is the only edit to the template's own stylesheet; everything else lives in `custom.css`
New features	—	Skip link, `<main>` landmark, back-to-top button, hover and focus transitions, reduced-motion support
Clean-up	Long SB Forms comment blocks, trailing slashes on void elements	Removed, which also clears the 17 “Info” messages from the W3C checker

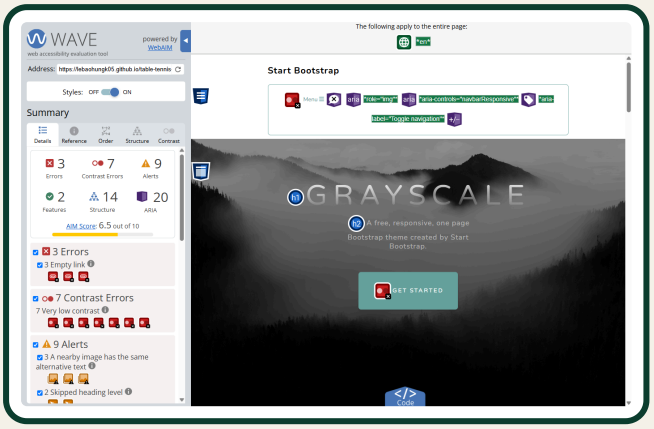
3. Accessibility Issues and How I Fixed Them

Before changing anything, I published the untouched template and ran it through WAVE, aXe and the W3C validators so I would know exactly what was broken. The template looks polished but fails several basic checks.

Original template compared with my site (same tools, both hosted on GitHub Pages)

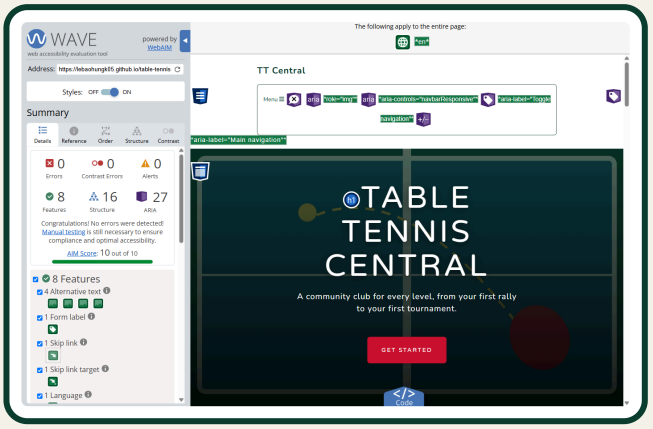
Check	Original	My site
WAVE errors	**3**	**0**
WAVE contrast errors	**7**	**0**
WAVE alerts	**9**	**0**
WAVE AIM score	**6.5 / 10**	**10 / 10**
aXe violations (axe-core 4.10.2)	**5 rules, 21 elements**	**0**
W3C HTML errors	**2 (plus 17 info)**	**0 (no messages at all)**
W3C CSS errors	**1**	**0**

ORIGINAL



WAVE on the original: 3 errors, 7 contrast errors, 9 alerts

MY SITE



WAVE on my site: “No errors were detected”

Each issue and its fix

Problems found in the template

Issue	Found by	Fix
Three social links contain only an icon, so screen readers announce “link” with no name	WAVE (Empty link), aXe (link-name)	Every icon link now has an `aria-label` that says where it goes

Issue	Found by	Fix
Headings jump from `h2` to `h4` in the Projects and Contact sections	W3C (2 errors), WAVE, aXe (heading-order)	Changed to `h3`, added an "Our Programs" `h2` and a visually hidden "Contact" `h2`, and turned the masthead tagline from a misused `h2` into a `<p>`
Low contrast: white text on teal buttons (about 2.9:1) and 50% black or white text classes	WAVE (7), aXe (5)	New button colour `#c8102e` with white text (5.9:1); replaced `text-black-50` and `text-white-50` with my own `text-soft-dark` (`#3b3b3b`) and `text-soft-light` classes
No `<main>` landmark, so ten blocks of content sit outside any landmark	aXe (landmark-one-main, region)	Wrapped every content section in `<main id="main-content">`, which is also the skip link's target
Images use `alt="..."`, which a screen reader reads as "dot dot dot"	WAVE (same alternative text alert)	Wrote a real description for every image
Email input has a placeholder but no visible label, and the button is permanently disabled	Manual testing	Added a visible `<label>`, enabled the button, and connected feedback with `aria-describedby`
Same-page links that point to `#!` go nowhere	WAVE (broken same-page link)	Removed or replaced with real destinations

Problems I ran into along the way

> ### My own CSS broke the mobile layout
>
> I gave the About image `max-width: 900px` to stop it growing too large on desktop. Because `custom.css` loads after Bootstrap, this replaced Bootstrap's `.img-fluid { max-width: 100% }`, and on a 390px phone the image became 812px wide and the

whole page scrolled sideways. The fix was `width: 100%; max-width: 900px;`. I then checked six widths from 360px to 1920px and none of them overflow.

WAVE still found one contrast error after my first pass

The visually hidden "Contact" heading I added inherited Bootstrap's dark body colour (`#212529`) inside the black contact section, so its contrast was 1.4:1. Nobody sees it, but a screen reader user with a custom stylesheet could. I made it white. I also gave every gradient section a solid `background-color` fallback, so text stays readable if the background image fails and checkers can measure the real contrast.

A specificity clash turned an icon the wrong colour

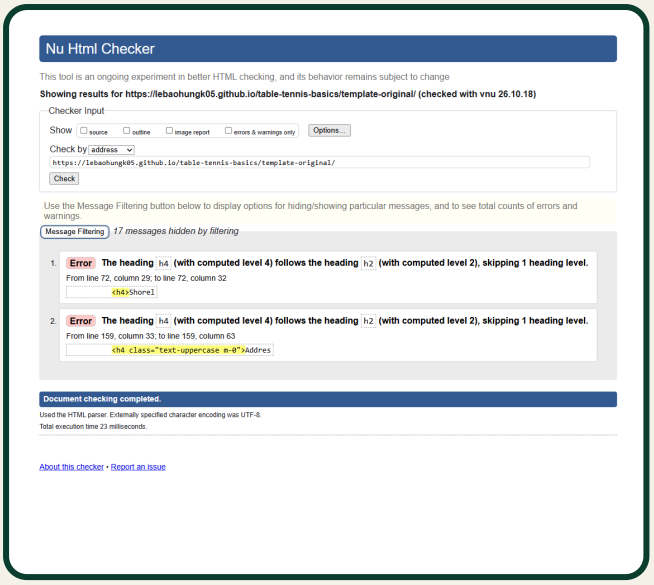
My `.footer a` rule (yellow links) was more specific than `.back-to-top`, so the back-to-top arrow turned yellow on red. I raised the selector to `.footer .back-to-top` so the arrow is white as intended.

4. W3C Validator Compliance

The final HTML passes the W3C Nu Html Checker with no errors, warnings or info messages. Both stylesheets pass the W3C CSS Validation Service with no errors. All screenshots are the validators' own result pages for the live URLs.

HTML

ORIGINAL



2 heading-level errors (the 17 info messages are hidden with the checker's own filter)

MY SITE



“No errors or warnings to show”

CSS

ORIGINAL STYLES.CSS



1 error: `fade(#fff, 80%)`

FIXED STYLES.CSS



No error found

MY CUSTOM.CSS



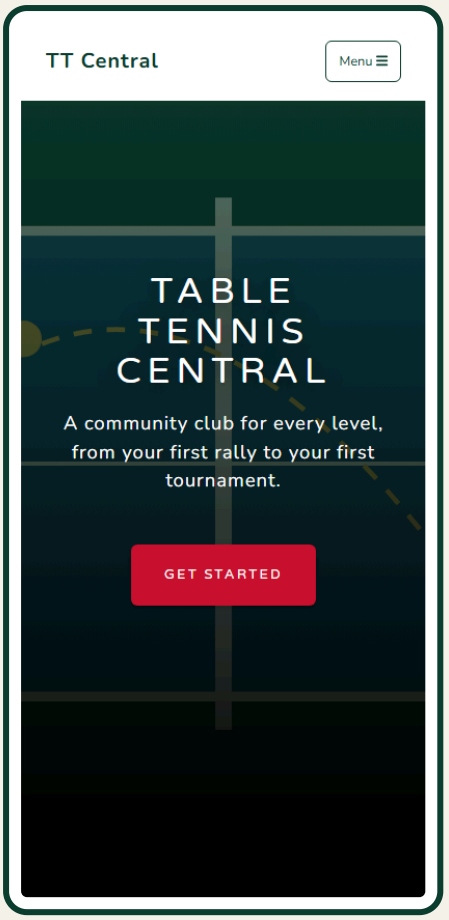
No error found

The CSS validator still lists warnings, which the course says are not deducted. `styles.css` has 130, all inherited from Bootstrap's vendor prefixes. `custom.css` has 5: four are the validator noting that it cannot statically check CSS variables, and one is the `-webkit-text-fill-color` reset I need to cancel the template's transparent heading text.

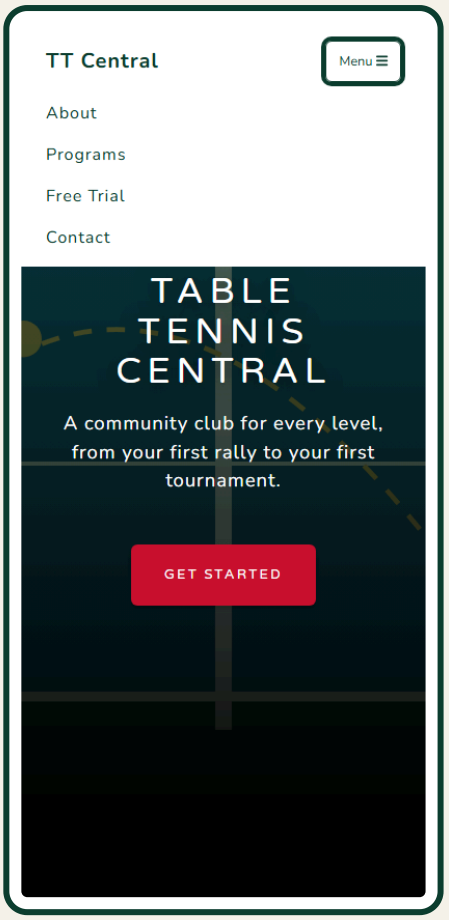
5. Responsive Design

The template's Bootstrap grid already handled the main column changes, and I kept it. On phones the menu collapses behind a “Menu” button, each program becomes an image stacked above its text, and the three contact cards stack. From 992px up the navigation

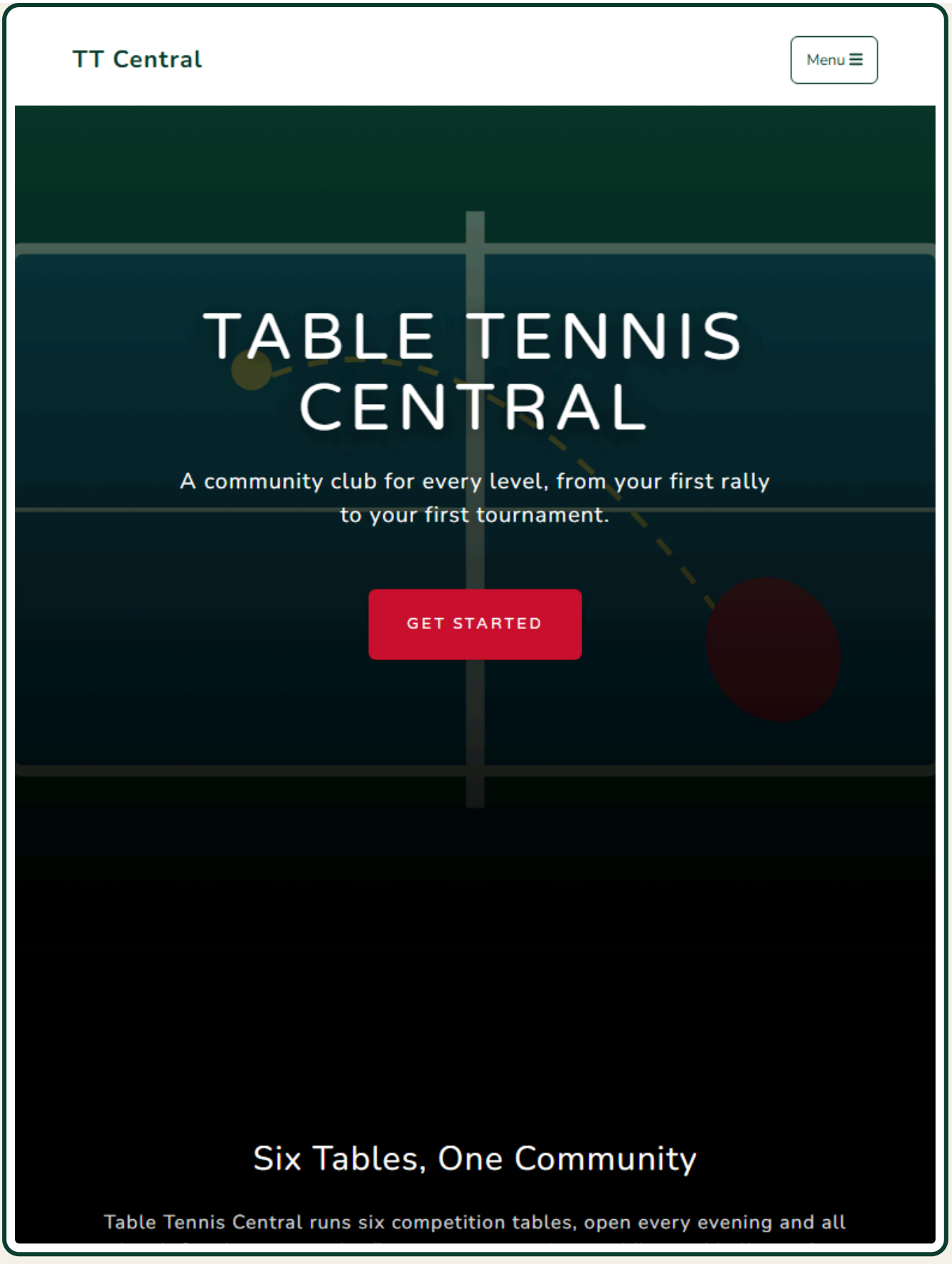
spreads out and programs sit beside their images. My additions had to respect that system. The title uses `clamp()` instead of fixed breakpoints, and nothing in `custom.css` sets a fixed width without a `max-width` or `%` fallback.



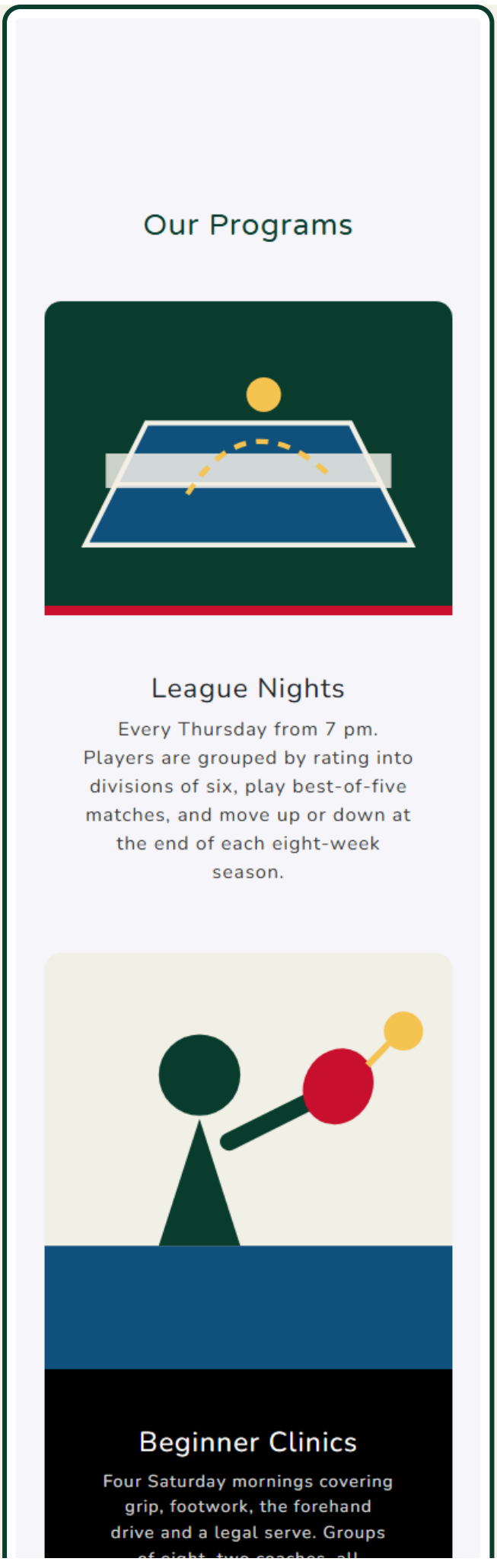
Phone · 390px

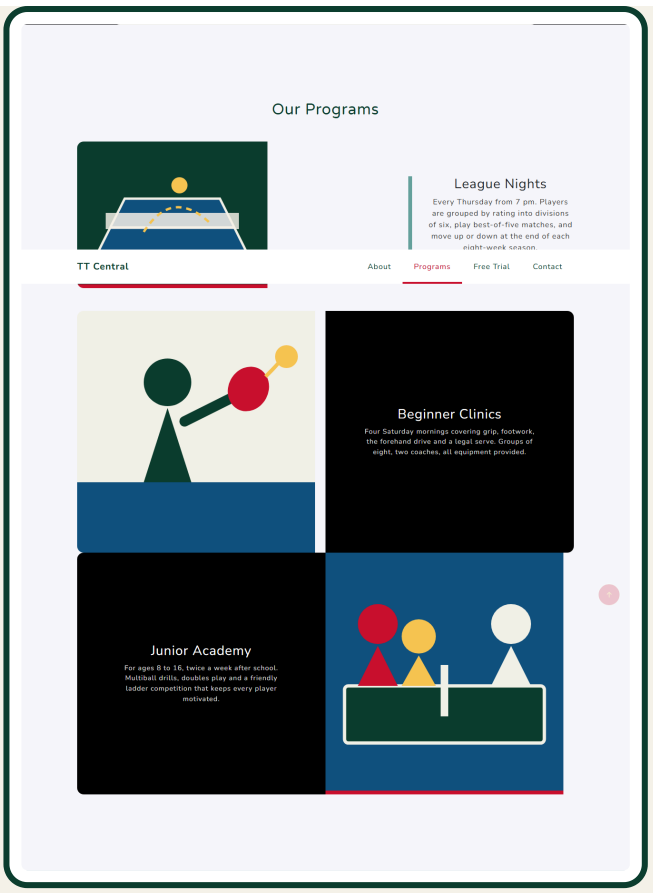


Phone menu open



Tablet · 768px





Programs on a desktop: side by side, alternating



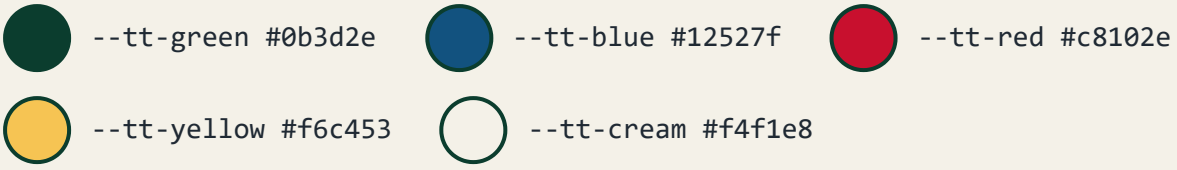
Programs on a phone: image above text

```
.masthead h1 {
  max-width: 14ch;
  font-size: clamp(2.2rem, 7vw, 5.5rem);
  line-height: 1.1;
  letter-spacing: 0.35rem;
}

.about-image {
  width: 100%;
  max-width: 900px;
}
```

6. Styling

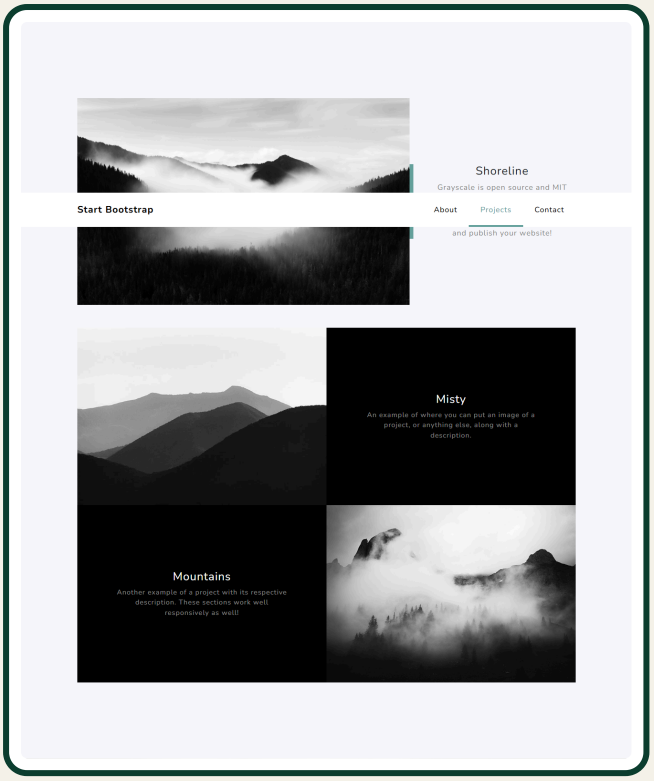
Instead of editing Bootstrap's 11,000-line stylesheet, all my styling lives in a separate `css/custom.css` that loads after it. A single block of CSS custom properties defines the palette, and every rule reads from those variables. Bootstrap 5 also exposes its own button colours as variables, so I re-themed the buttons by overriding those variables rather than rewriting the button rules.




```
:root {
  --tt-green: #0b3d2e;
  --tt-blue: #12527f;
  --tt-red: #c8102e;
  --tt-red-dark: #9e0c24;
  --tt-yellow: #f6c453;
  --tt-text-dark: #3b3b3b;
  --tt-text-light: rgba(255, 255, 255, 0.85);
  --tt-ease: 0.3s ease;
}

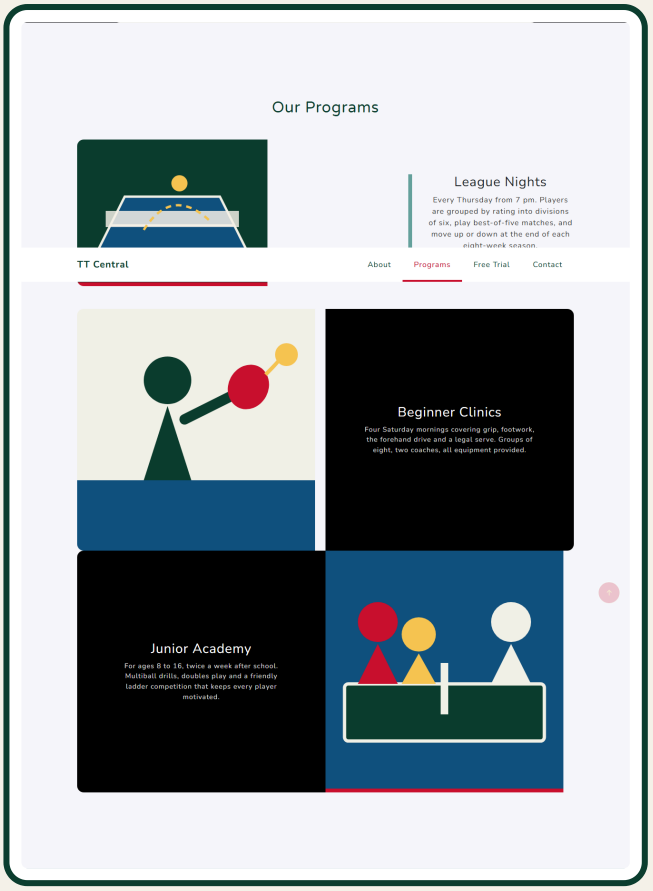
.btn-primary {
  --bs-btn-bg: var(--tt-red);
  --bs-btn-border-color: var(--tt-red);
  --bs-btn-hover-bg: var(--tt-red-dark);
  --bs-btn-active-bg: var(--tt-red-dark);
}
```

ORIGINAL



Template “Projects”

MY SITE



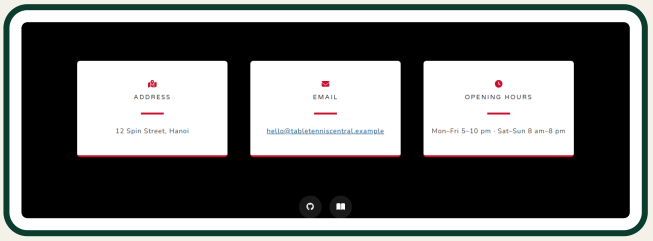
My “Programs”, with `:nth-of-type` accent strips

ORIGINAL



Template contact, with icon-only links

MY SITE



My contact, with labelled links

7. Four Extras

None of these four features exist in the original template. I designed and coded all of them.

1 Working, accessible sign-up form (JavaScript)

The template's form needs a paid service and a token, so out of the box it does nothing. I removed that dependency and wrote `js/signup.js`. The form checks for an empty or malformed email, marks the field with `aria-invalid`, moves focus back to it, and writes the message into a `role="status"` region so screen readers announce it. A valid address shows a confirmation and clears the field.



Invalid email: field outlined, message announced



Valid email: confirmation, field cleared

```
function handleSignup(event) {
  event.preventDefault();
  const input = document.getElementById("emailAddress");
  const status = document.getElementById("signupStatus");
  const email = input.value.trim();

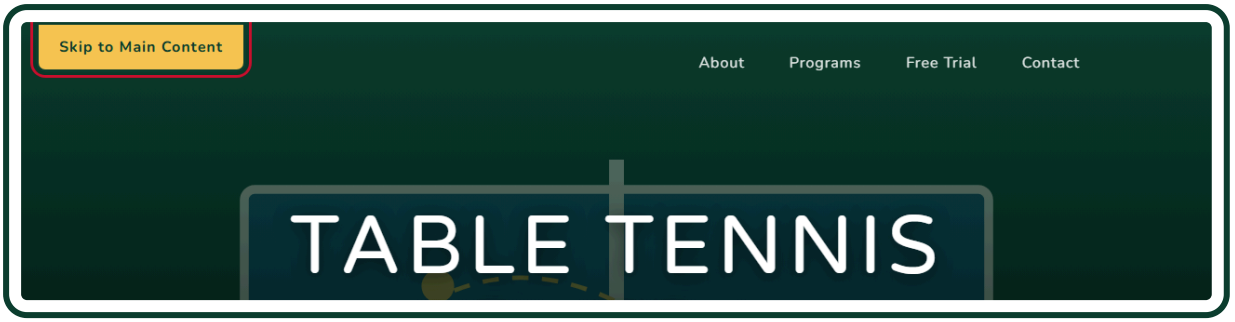
  if (!EMAIL_PATTERN.test(email)) {
    input.setAttribute("aria-invalid", "true");
    showStatus(status, "That email address does not look right. Try name@example
    input.focus();
    return;
  }

  input.removeAttribute("aria-invalid");
  showStatus(status, "Thanks! This week's open times are on their way to " + ema
  event.target.reset();
}
```

```
<label class="signup-label" for="emailAddress">Email address</label>
<input class="form-control" id="emailAddress" type="email" autocomplete="email"
       required aria-describedby="signupStatus">
<p class="signup-status" id="signupStatus" role="status" aria-live="polite"></p
```

2 Skip to Main Content link and landmark structure

The template has a fixed navigation bar that keyboard users must tab through on every visit. I added a skip link that stays hidden off screen until it receives focus, and wrapped the page content in a `<main>` landmark for it to jump to. That same landmark fixed aXe's two landmark violations.



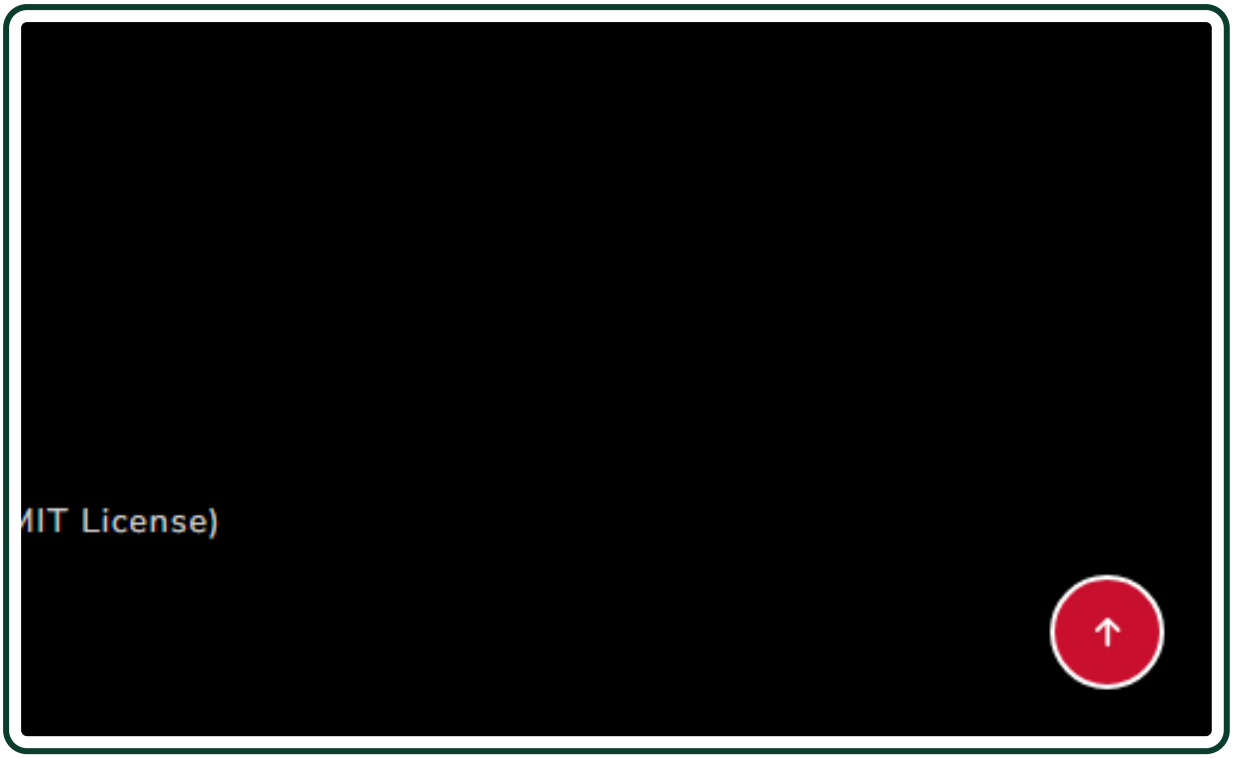
After pressing `Tab` once

```
.skip-link {
  position: absolute;
  top: -100px;
  left: 16px;
  z-index: 2000;
  transition: top 0.25s ease-in-out;
}

.skip-link:focus {
  top: 0;
  outline: 3px solid var(--tt-red);
}
```

3 Back-to-top button revealed on scroll (JavaScript)

On a long one-page site, a button in the corner saves scrolling back up. It stays invisible while the visitor is near the top, then fades and slides in after 600px of scrolling. When hidden it uses `visibility: hidden`, so keyboard users never tab onto an invisible control. It sits inside the footer landmark and has an `aria-label`.



Visible after scrolling down

```
function toggleBackToTop() {
  const button = document.getElementById("backToTop");
  button.classList.toggle("is-visible", window.scrollY > 600);
}

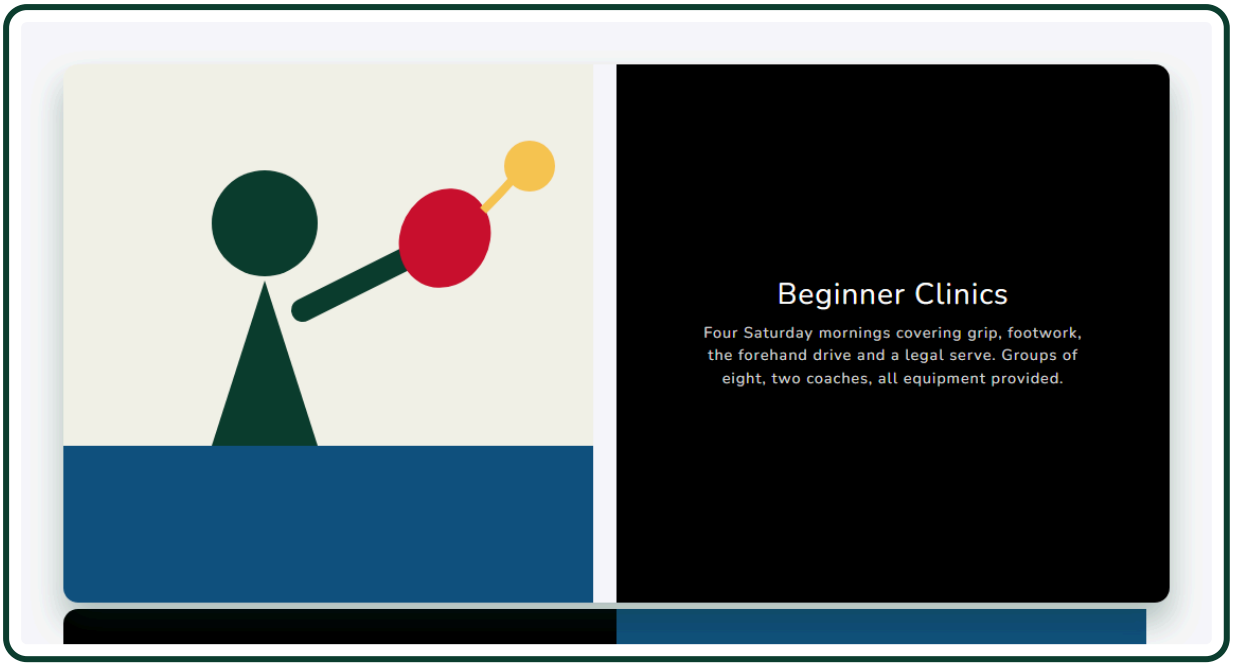
document.addEventListener("scroll", toggleBackToTop);
```

```
.footer .back-to-top {
  position: fixed;
  right: 20px;
  bottom: 20px;
  opacity: 0;
  visibility: hidden;
  transform: translateY(12px);
  transition: opacity var(--tt-ease), transform var(--tt-ease), visibility var(
}

.back-to-top.is-visible {
  opacity: 1;
  visibility: visible;
  transform: translateY(0);
}
```

4 Motion design that respects reduced-motion settings

Program cards lift and cast a shadow on hover, and they do the same on `:focus-within` so keyboard users get the effect too. Buttons rise slightly and links change colour, all with transitions. Some people get dizzy or nauseous from on-screen motion, so a `prefers-reduced-motion` media query turns every transition and transform off for visitors who have asked their operating system for less motion.



Program card on hover

```
.program-card {
  transition: transform var(--tt-ease), box-shadow var(--tt-ease);
}

.program-card:hover,
.program-card:focus-within {
  transform: translateY(-6px);
  box-shadow: 0 18px 30px rgba(11, 61, 46, 0.3);
}

@media (prefers-reduced-motion: reduce) {
  *,
  *::before,
  *::after {
    transition-duration: 0.01ms !important;
    scroll-behavior: auto !important;
  }

  .program-card:hover,
  .program-card:focus-within,
  .btn-primary:hover {
    transform: none;
  }
}
```